# LAPORAN PRAKTIKUM

# POSTTEST 3

# ALGORITMA PEMROGRAMAN LANJUT

**Disusun oleh:**

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**PROGRAM STUDI INFORMATIKA**

**UNIVERSITAS MULAWARMAN**

**SAMARINDA**

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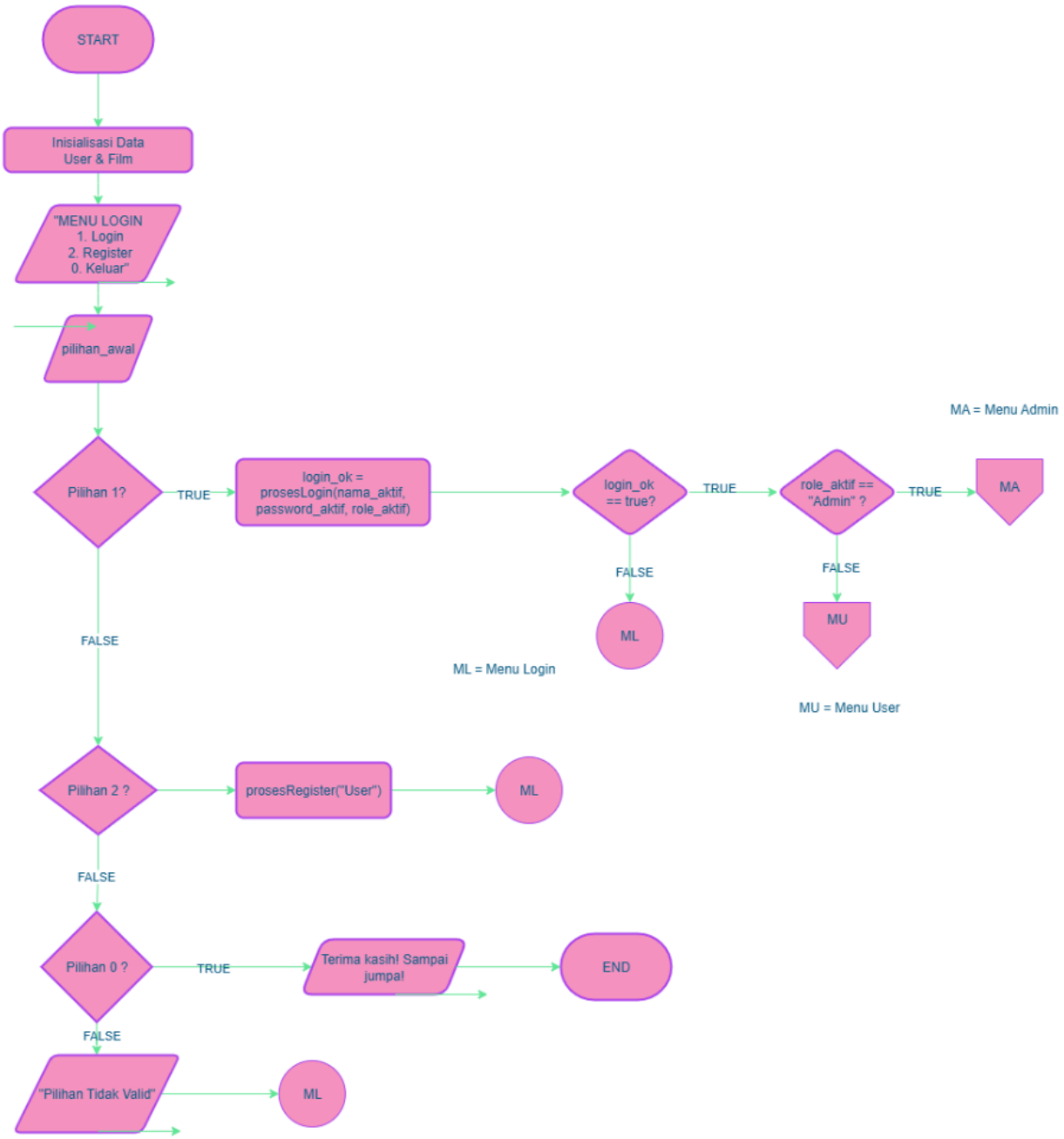
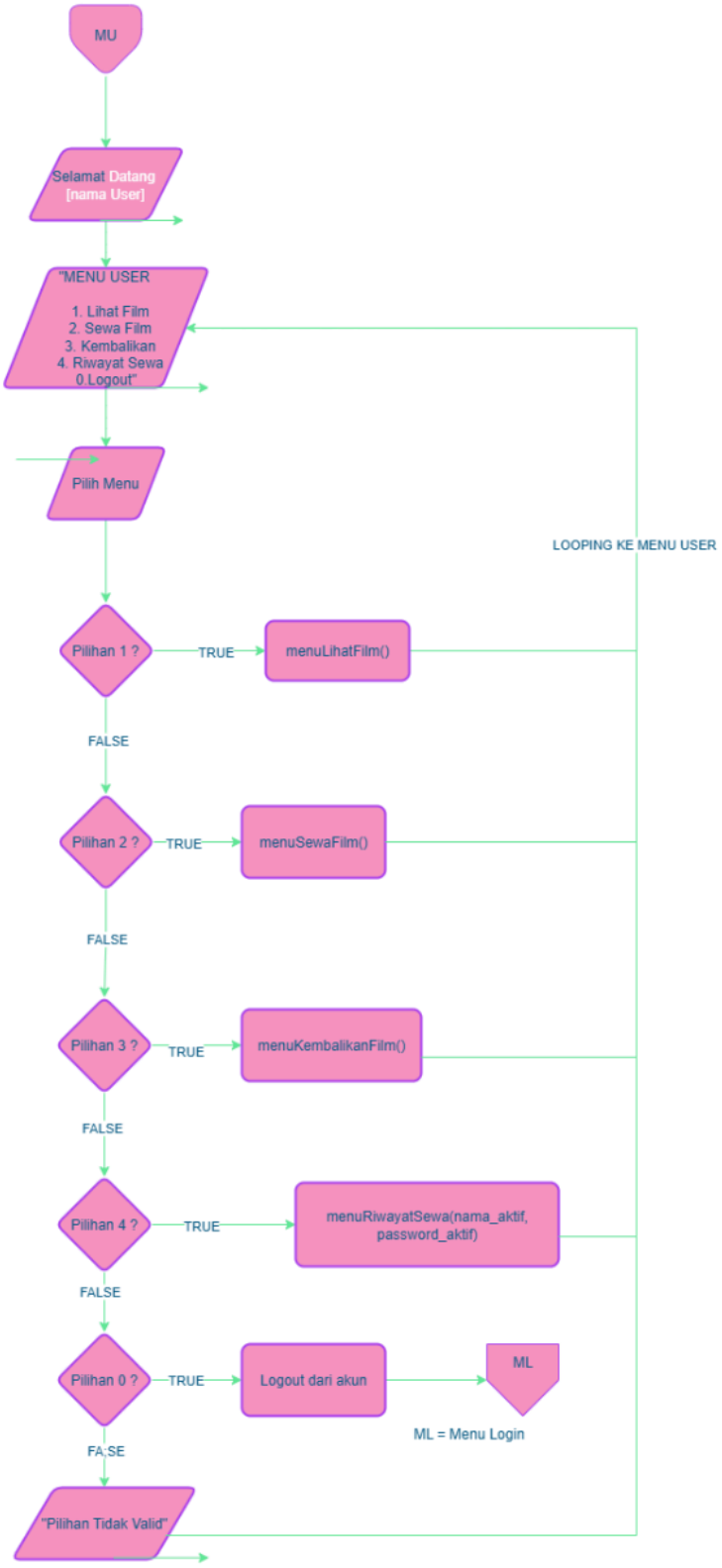
# 1. Flowchart

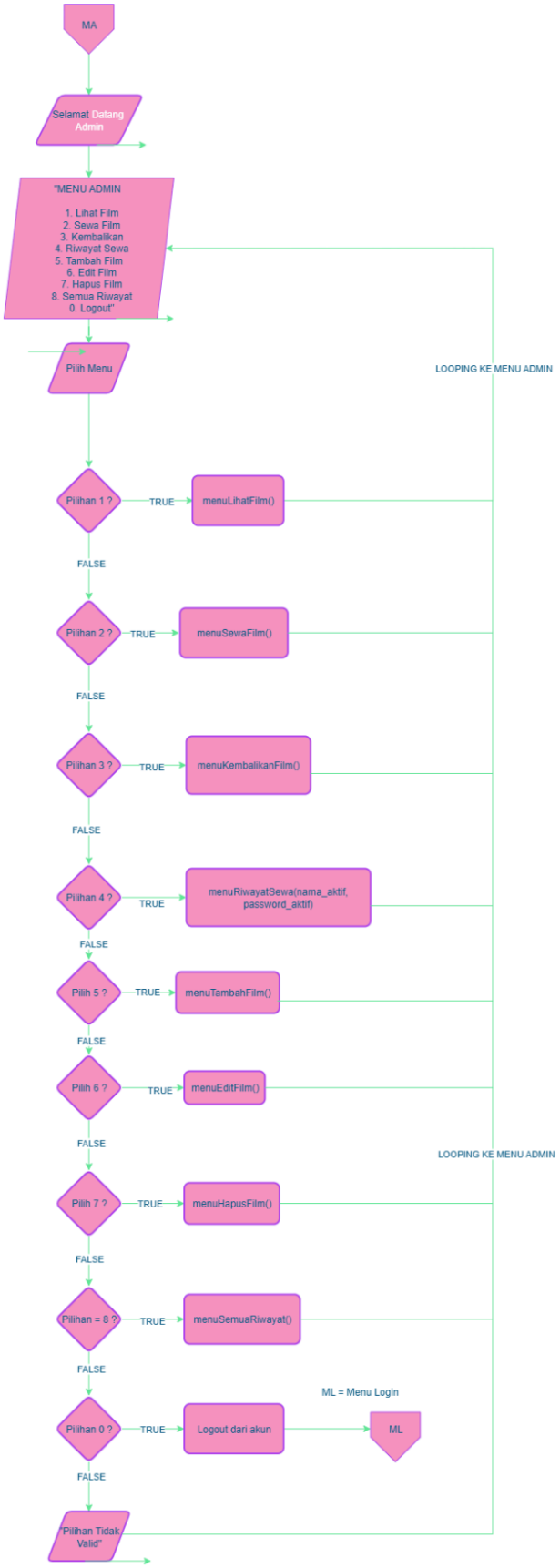
## 1.1 Gambar Bagian-Bagian Flowchart



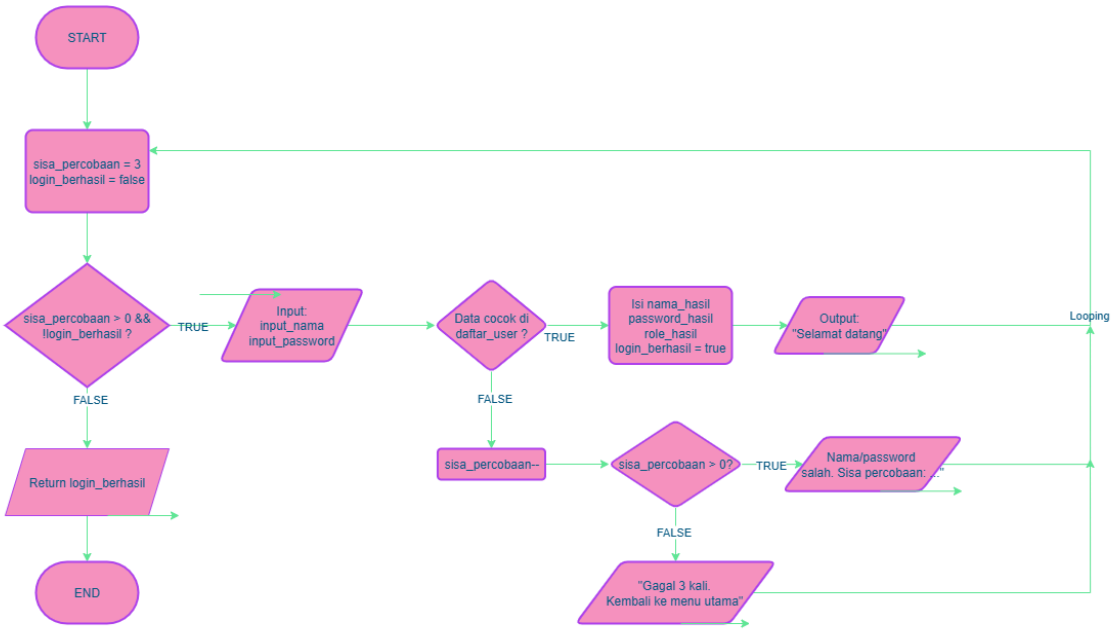

Gambar 2.1 Flowchart Bagian Menu Login



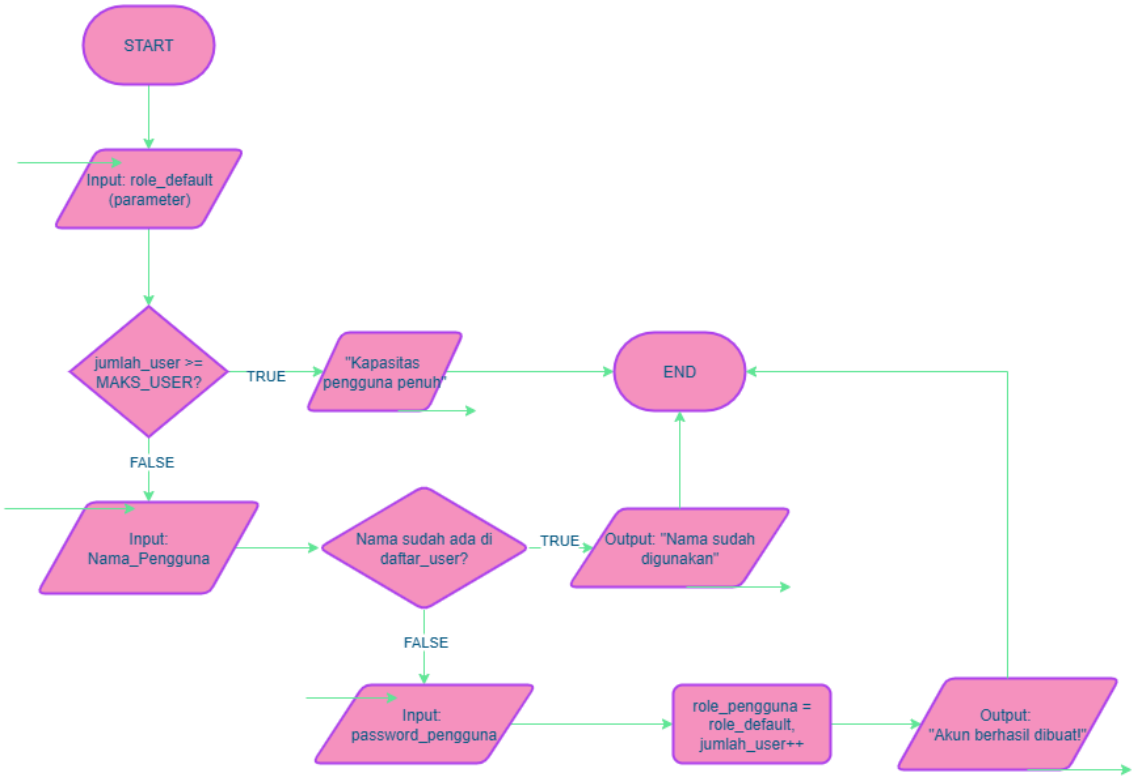

Gambar 2.2 Flowchart Bagian Menu User



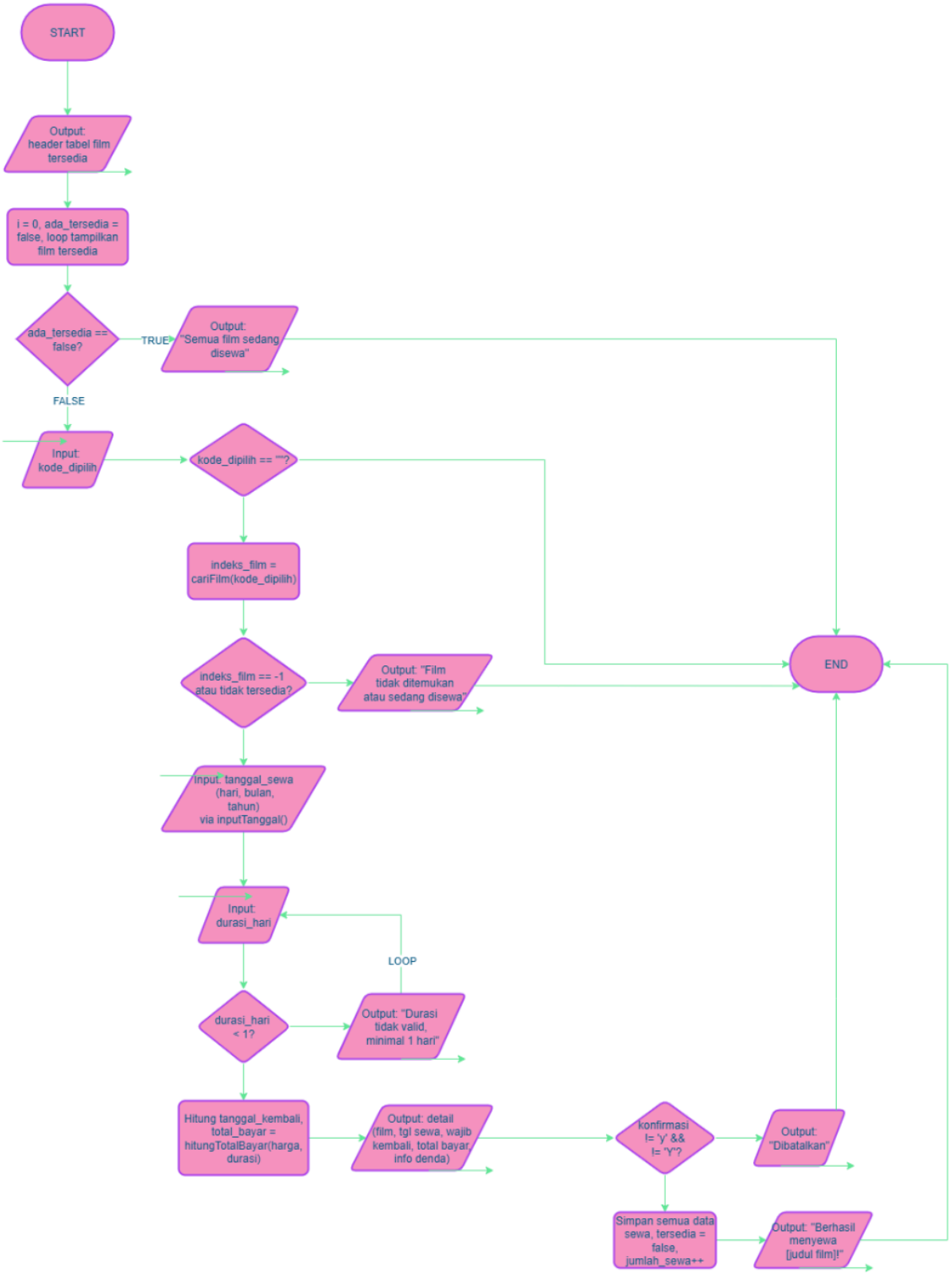

Gambar 2.3 Flowchart Bagian Menu Admin



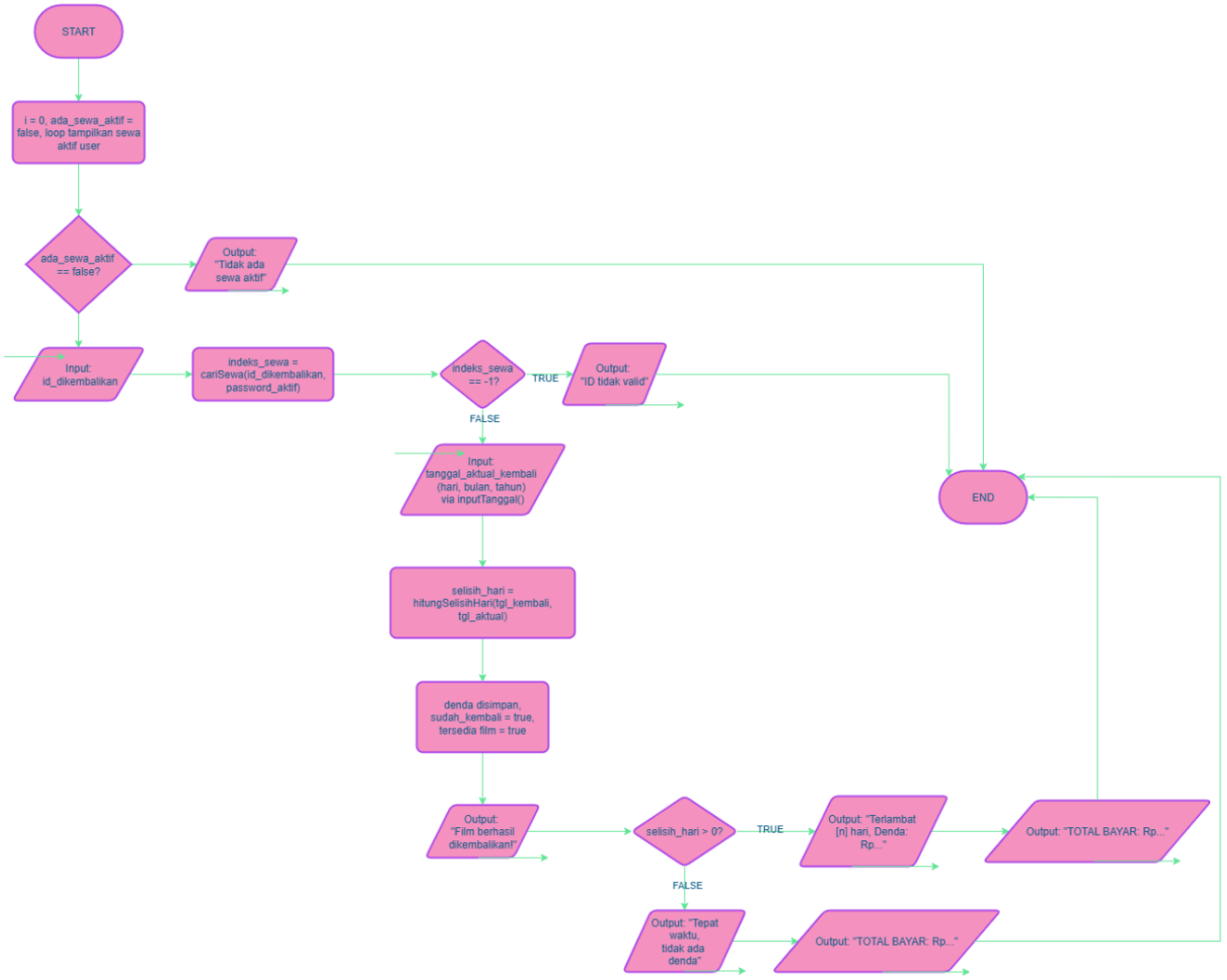

Gambar 2.4 Proses Login




Gambar 2.5 Proses Register




Gambar 2.6 Menu Sewa Film




Gambar 2.7 Menu Kembalikan Film

### 1.2 Penjelasan Flowchart

Program CinemaRent diawali dengan proses inisialisasi data film dan akun pengguna melalui prosedur inisialisasiData(), kemudian sistem menampilkan menu utama yang berisi pilihan login, register, dan keluar. Setiap fitur dalam program dipisahkan ke dalam sub-program berupa fungsi dan prosedur tersendiri agar struktur program lebih rapi, modular, dan mudah dipahami.

Proses pendaftaran akun ditangani oleh prosedur prosesRegister() yang menerima parameter role_default. Pada proses ini, sistem akan memeriksa apakah nama pengguna sudah digunakan sebelum menyimpan akun baru. Proses login ditangani oleh fungsi prosesLogin() yang menggunakan mekanisme pass by reference untuk menyimpan hasil autentikasi, dengan batas maksimal tiga kali percobaan. Jika login berhasil, sistem akan memeriksa role pengguna untuk menentukan apakah pengguna diarahkan ke Menu User atau Menu Admin.

Setelah berhasil masuk, program berjalan dalam bentuk perulangan hingga pengguna memilih logout. Setiap pilihan pada menu akan memanggil fungsi atau prosedur yang sesuai. Proses penyewaan film ditangani oleh menuSewaFilm() yang memanggil fungsi hitungTotalBayar() serta prosedur inputTanggal() untuk memastikan tanggal yang dimasukkan valid dan tidak melebihi 31 hari. Proses pengembalian film ditangani oleh menuKembalikanFilm() yang memanggil fungsi hitungSelisihHari() dan hitungDenda(). Fungsi hitungDenda() menerapkan konsep function overloading dengan dua versi parameter berbeda, serta memanfaatkan fungsi rekursif hitungDendaRekursif() untuk menghitung total denda berdasarkan jumlah hari keterlambatan.

Dalam flowchart, simbol terminator digunakan untuk menunjukkan awal dan akhir proses, simbol jajar genjang untuk input/output, belah ketupat untuk percabangan kondisi, dan persegi panjang untuk proses pengolahan data.

## 2. Deskripsi Singkat Program

Program CinemaRent merupakan aplikasi berbasis console yang dibuat menggunakan bahasa pemrograman C++. Program ini dirancang untuk membantu pengelolaan proses penyewaan film pada sebuah toko penyewaan bernama CinemaRent.

Pada Posttest 3, program dikembangkan dari versi sebelumnya dengan memisahkan setiap fitur ke dalam sub-program berupa fungsi dan prosedur yang terpisah dari fungsi utama main(). Seluruh fungsi dan prosedur menggunakan parameter, baik pass by value maupun pass by reference. Program juga menerapkan konsep function overloading pada fungsi hitungDenda(), yang memiliki dua versi berbeda yaitu hitungDenda(int selisih_hari) dan hitungDenda(int harga_per_hari, int selisih_hari). Selain itu, program menggunakan fungsi rekursif hitungDendaRekursif() untuk menghitung denda keterlambatan secara bertahap berdasarkan jumlah hari keterlambatan.

Program ini menyediakan dua jenis akses, yaitu User dan Admin. Setelah berhasil login, pengguna dapat melihat daftar film yang tersedia, menyewa film untuk jangka waktu tertentu dengan validasi tanggal, mengembalikan film yang dipinjam dengan perhitungan denda otomatis jika terjadi keterlambatan, serta melihat riwayat penyewaan. Admin memiliki hak akses tambahan untuk menambah, mengedit, dan menghapus data film, serta melihat seluruh riwayat penyewaan dari semua pengguna.

## 3. Source Code

### 3.1 Deklarasi Struct & Variabel Global

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

struct Studio {
    string nama_studio;
    string negara_studio;
};

struct Film {
    string kode_film;
    string judul_film;
    string genre_film;
    int    tahun_rilis;
    int    harga_sewa;
    Studio studio;
    bool   tersedia;
};

struct Pengguna {
    string nama_pengguna;
    string password_pengguna;
    string role_pengguna;
};

struct Tanggal {
    int hari;
    int bulan;
    int tahun;
};

struct DataSewa {
    int     id_sewa;
    string  password_penyewa;
    string  nama_penyewa;
    string  kode_film_sewa;
    string  judul_film_sewa;
    int     harga_per_hari;
    int     durasi_hari;
    Tanggal tanggal_sewa;
    Tanggal tanggal_kembali;
    int     total_bayar;
    int     denda;
```

```
    bool    sudah_kembali;
};

#define MAKS_FILM      50
#define MAKS_USER      20
#define MAKS_SEWA     100
#define DENDA_PER_HARI 5000

Film     daftar_film[MAKS_FILM];
int      jumlah_film = 0;

Pengguna daftar_user[MAKS_USER];
int      jumlah_user = 0;

DataSewa daftar_sewa[MAKS_SEWA];
int      jumlah_sewa   = 0;
int      id_berikutnya = 1;

string nama_aktif     = "";
string password_aktif = "";
string role_aktif     = "";
```

### 3.2 Deklarasi Fungsi & Prosedur

```
void inisialisasiData();
void menuLihatFilm();
void menuSewaFilm();
void menuKembalikanFilm();
void menuRiwayatSewa(string nama_pengguna, string password_pengguna);
void menuTambahFilm();
void menuEditFilm();
void menuHapusFilm();
void menuSemuaRiwayat();
void prosesRegister(string role_default);
void inputTanggal(int &hari, int &bulan, int &tahun);

bool prosesLogin(string &nama_hasil, string &password_hasil, string
&role_hasil);
int  hitungTotalBayar(int harga_per_hari, int durasi_hari);
int  hitungDendaRekursif(int selisih_hari);
int  hitungDenda(int selisih_hari);
int  hitungDenda(int harga_per_hari, int selisih_hari);
int  cariFilm(string kode_film);
int  cariSewa(int id_sewa, string password_penyewa);
int  hitungSelisihHari(Tanggal tanggal_wajib, Tanggal tanggal_aktual);
```

### 3.3 Fitur Inisialisasi Data

```
void inisialisasiData() {
    daftar_user[0] = {"Admin", "000",         "Admin"};
    daftar_user[1] = {"Nurul", "2509106033", "User"};
    jumlah_user = 2;

    daftar_film[0]  = {"F001", "The Nun II",                "Horror",    2023,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[1]  = {"F002", "Hereditary",                "Horror",    2018,
12000, {"A24",         "Amerika"}, true};
    daftar_film[2]  = {"F003", "It (2017)",                 "Horror",    2017,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[3]  = {"F004", "Talk to Me",                "Horror",    2023,
12000, {"A24",         "Amerika"}, true};
    daftar_film[4]  = {"F005", "Your Name",                 "Romance",   2016,
10000, {"Toho",        "Jepang"},  true};
    daftar_film[5]  = {"F006", "To All The Boys",           "Romance",   2018,
10000, {"Netflix",     "Amerika"}, true};
    daftar_film[6]  = {"F007", "Five Feet Apart",           "Romance",   2019,
10000, {"Warner Bros", "Amerika"}, true};
    daftar_film[7]  = {"F008", "A Silent Voice",            "Romance",   2016,
10000, {"Toho",        "Jepang"},  true};
    daftar_film[8]  = {"F009", "The Dark Knight",           "Action",    2008,
13000, {"Warner Bros", "Amerika"}, true};
    daftar_film[9]  = {"F010", "John Wick",                 "Action",    2014,
13000, {"Lionsgate",   "Amerika"}, true};
    daftar_film[10] = {"F011", "Mad Max: Fury Road",        "Action",    2015,
13000, {"Warner Bros", "Amerika"}, true};
    daftar_film[11] = {"F012", "Demon Slayer: Mugen",       "Action",    2020,
13000, {"Toho",        "Jepang"},  true};
    daftar_film[12] = {"F013", "Saltburn",                  "Thriller",  2023,
12000, {"A24",         "Amerika"}, true};
    daftar_film[13] = {"F014", "Gone Girl",                 "Thriller",  2014,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[14] = {"F015", "Bird Box",                  "Thriller",  2018,
12000, {"Netflix",     "Amerika"}, true};
    daftar_film[15] = {"F016", "Midsommar",                 "Thriller",  2019,
12000, {"A24",         "Amerika"}, true};
    daftar_film[16] = {"F017", "Super Mario Bros Movie",    "Comedy",    2023,
10000, {"Nintendo",    "Amerika"}, true};
    daftar_film[17] = {"F018", "Puss in Boots 2",           "Comedy",    2022,
10000, {"DreamWorks",  "Amerika"}, true};
    daftar_film[18] = {"F019", "How To Train Your Dragon", "Animation", 2014,
10000, {"DreamWorks",  "Amerika"}, true};
    daftar_film[19] = {"F020", "Spirited Away",             "Animation", 2001,
10000, {"Toho",        "Jepang"},  true};
    jumlah_film = 20;
}
```

### 3.4 Fitur Login

```
bool prosesLogin(string &nama_hasil, string &password_hasil, string &role_hasil)
{
    int  sisa_percobaan = 3;
    bool login_berhasil = false;

    while (sisa_percobaan > 0 && !login_berhasil) {
        string input_nama, input_password;
        cin.ignore();
        cout << "  Nama      : "; getline(cin, input_nama);
        cout << "  Password  : "; getline(cin, input_password);
        for (int i = 0; i < jumlah_user; i++) {
            if (daftar_user[i].nama_pengguna     == input_nama &&
                daftar_user[i].password_pengguna == input_password) {
                nama_hasil     = daftar_user[i].nama_pengguna;
                password_hasil = daftar_user[i].password_pengguna;
                role_hasil     = daftar_user[i].role_pengguna;
                login_berhasil = true;
                break;
            }
        }
        if (login_berhasil) {
            cout << "  [v] Selamat datang, " << nama_hasil << "!" << endl;
            cout << "  [Tekan Enter...]"; cin.get();
        } else {
            sisa_percobaan--;
            if (sisa_percobaan > 0)
                cout << "  [!] Nama/password salah. Sisa percobaan: " <<
sisa_percobaan << endl;
            else
                cout << "  [!!!] Gagal 3 kali. Kembali ke menu utama." << endl;
        }
    }
    return login_berhasil;
}
```

### 3.5 Fitur Register

```
void prosesRegister(string role_default) {
    if (jumlah_user >= MAKS_USER) {
        cout << "  [!] Kapasitas pengguna penuh." << endl;
        cout << "  [Tekan Enter...]"; cin.ignore(); cin.get(); return;
    }
    cin.ignore();
    cout << "  Nama      : "; getline(cin,
```

```
daftar_user[jumlah_user].nama_pengguna);
    bool nama_sudah_ada = false;
    for (int i = 0; i < jumlah_user; i++)
        if (daftar_user[i].nama_pengguna ==
daftar_user[jumlah_user].nama_pengguna) {
            nama_sudah_ada = true; break;
        }
    if (nama_sudah_ada) {
        cout << "  [!] Nama sudah digunakan." << endl;
        daftar_user[jumlah_user].nama_pengguna = "";
    } else {
        cout << "  Password  : "; getline(cin,
daftar_user[jumlah_user].password_pengguna);
        daftar_user[jumlah_user].role_pengguna = role_default;
        jumlah_user++;
        cout << "  [v] Akun berhasil dibuat!" << endl;
    }
    cout << "  [Tekan Enter...]"; cin.get();
}
```

### 3.6 Fitur Lihat Film

```
void menuLihatFilm() {
    cout <<
"+-------+-----------------------------+-----------+-------+----------+--------
-+" << endl;
    cout << "| Kode  | Judul                         | Genre     | Tahun |
Harga/Hr | Status  |" << endl;
    cout <<
"+-------+-----------------------------+-----------+-------+----------+--------
-+" << endl;
    for (int i = 0; i < jumlah_film; i++) {
        string status_film = daftar_film[i].tersedia ? "Tersedia" : "Disewa  ";
        string harga_str   = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << "| " << left << setw(5)  << daftar_film[i].kode_film
             << "| "         << setw(29) << daftar_film[i].judul_film
             << "| "         << setw(10) << daftar_film[i].genre_film
             << "| "         << setw(5)  << daftar_film[i].tahun_rilis
             << " | "        << setw(8)  << harga_str
             << "| "         << status_film << "|" << endl;
    }
    cout <<
"+-------+-----------------------------+-----------+-------+----------+--------
-+" << endl;
    cout << "  [Tekan Enter...]"; cin.ignore(); cin.get();
}
```

### 3.7 Fitur Sewa Film

```
void inputTanggal(int &hari, int &bulan, int &tahun) {
    do {
        cout << "  Hari  (1-31) : "; cin >> hari;
        if (hari < 1 || hari > 31) cout << "  [!] Hari tidak valid, masukkan
1-31." << endl;
    } while (hari < 1 || hari > 31);

    do {
        cout << "  Bulan (1-12) : "; cin >> bulan;
        if (bulan < 1 || bulan > 12) cout << "  [!] Bulan tidak valid, masukkan
1-12." << endl;
    } while (bulan < 1 || bulan > 12);

    do {
        cout << "  Tahun        : "; cin >> tahun;
        if (tahun < 2000) cout << "  [!] Tahun tidak valid, masukkan >= 2000."
<< endl;
    } while (tahun < 2000);
}
int cariFilm(string kode_film) {
    for (int i = 0; i < jumlah_film; i++) {
        if (daftar_film[i].kode_film == kode_film) return i;
    }
    return -1;
}

int hitungTotalBayar(int harga_per_hari, int durasi_hari) {
    return harga_per_hari * durasi_hari;
}
void menuSewaFilm() {
    cout << "  === SEWA FILM ===" << endl;
    cout << "+-------+-----------------------------+-----------+----------+" <<
endl;
    cout << "| Kode  | Judul                       | Genre     | Harga/Hr |" <<
endl;
    cout << "+-------+-----------------------------+-----------+----------+" <<
endl;

    bool ada_tersedia = false;
    for (int i = 0; i < jumlah_film; i++) {
        if (!daftar_film[i].tersedia) continue;
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << "| " << left << setw(5)  << daftar_film[i].kode_film
             << "| "         << setw(29) << daftar_film[i].judul_film
             << "| "         << setw(10) << daftar_film[i].genre_film
             << "| "         << setw(8)  << harga_str << "|" << endl;
        ada_tersedia = true;
    }
```

```
    cout << "+-------+------------------------------+-----------+----------+" <<
endl;

    if (!ada_tersedia) {
        cout << "  Semua film sedang disewa." << endl;
        cin.ignore(); cin.get(); return;
    }

    string kode_dipilih;
    cout << "  Kode film (Enter -> Kembali) : ";
    cin.ignore(); getline(cin, kode_dipilih);
    if (kode_dipilih == "") return;

    int indeks_film = cariFilm(kode_dipilih);
    if (indeks_film == -1 || !daftar_film[indeks_film].tersedia) {
        cout << "  [!] Film tidak ditemukan atau sedang disewa." << endl;
        cin.get(); return;
    }

    Tanggal tanggal_sewa;
    int durasi_hari;
    cout << "  Tanggal Sewa:" << endl;
    inputTanggal(tanggal_sewa.hari, tanggal_sewa.bulan, tanggal_sewa.tahun);
    do {
        cout << "  Durasi (hari, min 1) : "; cin >> durasi_hari;
        if (durasi_hari < 1) cout << "  [!] Durasi tidak valid, minimal 1 hari."
<< endl;
    } while (durasi_hari < 1);

    Tanggal tanggal_kembali = tanggal_sewa;
    int hari_per_bulan[13]  = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30,
31};
    tanggal_kembali.hari   += durasi_hari;
    while (tanggal_kembali.hari > hari_per_bulan[tanggal_kembali.bulan]) {
        tanggal_kembali.hari -= hari_per_bulan[tanggal_kembali.bulan];
        tanggal_kembali.bulan++;
        if (tanggal_kembali.bulan > 12) { tanggal_kembali.bulan = 1;
tanggal_kembali.tahun++; }
    }

    int total_bayar_sewa = hitungTotalBayar(daftar_film[indeks_film].harga_sewa,
durasi_hari);

    cout << "  Film          : " << daftar_film[indeks_film].judul_film << endl;
    cout << "  Tgl Sewa      : ";
    if (tanggal_sewa.hari   < 10) cout << "0"; cout << tanggal_sewa.hari   <<
"/";
    if (tanggal_sewa.bulan  < 10) cout << "0"; cout << tanggal_sewa.bulan  <<
"/" << tanggal_sewa.tahun  << endl;
    cout << "  Wajib Kembali : ";
    if (tanggal_kembali.hari  < 10) cout << "0"; cout << tanggal_kembali.hari
```

```
<< "/";
    if (tanggal_kembali.bulan < 10) cout << "0"; cout << tanggal_kembali.bulan
<< "/" << tanggal_kembali.tahun << endl;
    cout << "  Total Bayar   : Rp" << total_bayar_sewa << endl;
    cout << "  Denda         : Rp" << DENDA_PER_HARI << "/hari jika terlambat"
<< endl;
    cout << "  Konfirmasi? (y/n) : ";
    char konfirmasi_sewa; cin >> konfirmasi_sewa;
    if (konfirmasi_sewa != 'y' && konfirmasi_sewa != 'Y') {
        cout << "  Dibatalkan." << endl; cin.ignore(); cin.get(); return;
    }

    daftar_film[indeks_film].tersedia         = false;
    daftar_sewa[jumlah_sewa].id_sewa          = id_berikutnya++;
    daftar_sewa[jumlah_sewa].password_penyewa = password_aktif;
    daftar_sewa[jumlah_sewa].nama_penyewa     = nama_aktif;
    daftar_sewa[jumlah_sewa].kode_film_sewa   = kode_dipilih;
    daftar_sewa[jumlah_sewa].judul_film_sewa  =
daftar_film[indeks_film].judul_film;
    daftar_sewa[jumlah_sewa].harga_per_hari   =
daftar_film[indeks_film].harga_sewa;
    daftar_sewa[jumlah_sewa].durasi_hari      = durasi_hari;
    daftar_sewa[jumlah_sewa].tanggal_sewa     = tanggal_sewa;
    daftar_sewa[jumlah_sewa].tanggal_kembali  = tanggal_kembali;
    daftar_sewa[jumlah_sewa].total_bayar      = total_bayar_sewa;
    daftar_sewa[jumlah_sewa].denda            = 0;
    daftar_sewa[jumlah_sewa].sudah_kembali    = false;
    jumlah_sewa++;

    cout << "  [v] Berhasil menyewa \"" << daftar_film[indeks_film].judul_film
<< "\"!" << endl;
    cout << "  [Tekan Enter...]"; cin.ignore(); cin.get();
}
```

### 3.8 Fitur Kembalikan Film

```
int hitungSelisihHari(Tanggal tanggal_wajib, Tanggal tanggal_aktual) {
    int total_wajib  = tanggal_wajib.tahun  * 365 + tanggal_wajib.bulan  * 30 +
tanggal_wajib.hari;
    int total_aktual = tanggal_aktual.tahun * 365 + tanggal_aktual.bulan * 30 +
tanggal_aktual.hari;
    return total_aktual - total_wajib;
}
int hitungDendaRekursif(int selisih_hari) {
    if (selisih_hari <= 0) return 0;
    return DENDA_PER_HARI + hitungDendaRekursif(selisih_hari - 1);
}
int hitungDenda(int selisih_hari) {
```

```
    if (selisih_hari <= 0) return 0;
    return hitungDendaRekursif(selisih_hari);
}
int hitungDenda(int harga_per_hari, int selisih_hari) {
    if (selisih_hari <= 0) return 0;
    return selisih_hari * harga_per_hari;
}
int cariSewa(int id_sewa, string password_penyewa) {
    for (int i = 0; i < jumlah_sewa; i++) {
        if (daftar_sewa[i].id_sewa == id_sewa &&
            daftar_sewa[i].password_penyewa == password_penyewa &&
            !daftar_sewa[i].sudah_kembali) return i;
    }
    return -1;
}
void menuKembalikanFilm() {
    cout << "  === KEMBALIKAN FILM ===" << endl;
    cout << "+----+------------------------------+------------+------------+" <<
endl;
    cout << "| ID | Judul                        | Tgl Sewa   | Wajib Kemb |" <<
endl;
    cout << "+----+------------------------------+------------+------------+" <<
endl;

    bool ada_sewa_aktif = false;
    for (int i = 0; i < jumlah_sewa; i++) {
        if (daftar_sewa[i].password_penyewa != password_aktif ||
daftar_sewa[i].sudah_kembali) continue;

        string tgl_sewa_str = "";
        if (daftar_sewa[i].tanggal_sewa.hari   < 10) tgl_sewa_str += "0";
        tgl_sewa_str += to_string(daftar_sewa[i].tanggal_sewa.hari) + "/";
        if (daftar_sewa[i].tanggal_sewa.bulan  < 10) tgl_sewa_str += "0";
        tgl_sewa_str += to_string(daftar_sewa[i].tanggal_sewa.bulan) + "/" +
to_string(daftar_sewa[i].tanggal_sewa.tahun);

        string tgl_kemb_str = "";
        if (daftar_sewa[i].tanggal_kembali.hari   < 10) tgl_kemb_str += "0";
        tgl_kemb_str += to_string(daftar_sewa[i].tanggal_kembali.hari) + "/";
        if (daftar_sewa[i].tanggal_kembali.bulan  < 10) tgl_kemb_str += "0";
        tgl_kemb_str += to_string(daftar_sewa[i].tanggal_kembali.bulan) + "/" +
to_string(daftar_sewa[i].tanggal_kembali.tahun);

        cout << "| " << left << setw(2)  << daftar_sewa[i].id_sewa
             << " | "        << setw(29) << daftar_sewa[i].judul_film_sewa
             << "| "         << setw(11) << tgl_sewa_str
             << "| "         << setw(11) << tgl_kemb_str << "|" << endl;
        ada_sewa_aktif = true;
    }
    cout << "+----+------------------------------+------------+------------+" <<
endl;
```

```
    if (!ada_sewa_aktif) {
        cout << "  Tidak ada sewa aktif." << endl; cin.ignore(); cin.get();
return;
    }

    int id_dikembalikan;
    cout << "  ID sewa : "; cin >> id_dikembalikan;

    int indeks_sewa = cariSewa(id_dikembalikan, password_aktif);
    if (indeks_sewa == -1) {
        cout << "  [!] ID tidak valid." << endl; cin.ignore(); cin.get();
return;
    }

    Tanggal tanggal_aktual_kembali;
    cout << "  Tanggal Kembali:" << endl;
    inputTanggal(tanggal_aktual_kembali.hari, tanggal_aktual_kembali.bulan,
tanggal_aktual_kembali.tahun);

    int selisih_hari    =
hitungSelisihHari(daftar_sewa[indeks_sewa].tanggal_kembali,
tanggal_aktual_kembali);
    int denda_terlambat = hitungDenda(selisih_hari);

    daftar_sewa[indeks_sewa].denda         = denda_terlambat;
    daftar_sewa[indeks_sewa].sudah_kembali = true;

    int indeks_film_kembali = cariFilm(daftar_sewa[indeks_sewa].kode_film_sewa);
    if (indeks_film_kembali != -1) daftar_film[indeks_film_kembali].tersedia =
true;

    cout << "  [v] Film berhasil dikembalikan!" << endl;
    if (selisih_hari > 0) {
        cout << "  Terlambat " << selisih_hari << " hari" << endl;
        cout << "  Denda     : Rp" << denda_terlambat << endl;
    } else {
        cout << "  Tepat waktu, tidak ada denda." << endl;
    }
    cout << "  TOTAL BAYAR: Rp" << daftar_sewa[indeks_sewa].total_bayar +
denda_terlambat << endl;
    cout << "  [Tekan Enter...]"; cin.ignore(); cin.get();
}
```

### 3.9 Fitur Tambah Film (Admin)

```
void menuTambahFilm() {
    cout << "  === TAMBAH FILM ===" << endl;
    if (jumlah_film >= MAKS_FILM) {
        cout << "  [!] Kapasitas penuh." << endl; cin.ignore(); cin.get();
return;
    }
    cout <<
"+-------+-----------------------------+-----------+-------+----------+" <<
endl;
    cout << "| Kode  | Judul                         | Genre     | Tahun |
Harga/Hr |" << endl;
    cout <<
"+-------+-----------------------------+-----------+-------+----------+" <<
endl;
    for (int i = 0; i < jumlah_film; i++) {
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << "| " << left << setw(5)  << daftar_film[i].kode_film
             << "| "         << setw(29) << daftar_film[i].judul_film
             << "| "         << setw(10) << daftar_film[i].genre_film
             << "| "         << setw(5)  << daftar_film[i].tahun_rilis
             << " | "        << setw(8)  << harga_str << "|" << endl;
    }
    cout <<
"+-------+-----------------------------+-----------+-------+----------+" <<
endl;
    cin.ignore();
    cout << "  Kode   : "; getline(cin, daftar_film[jumlah_film].kode_film);
    if (daftar_film[jumlah_film].kode_film == "") return;

    bool kode_duplikat = false;
    for (int i = 0; i < jumlah_film; i++)
        if (daftar_film[i].kode_film == daftar_film[jumlah_film].kode_film) {
kode_duplikat = true; break; }
    if (kode_duplikat) {
        cout << "  [!] Kode sudah ada." << endl;
        daftar_film[jumlah_film].kode_film = ""; cin.get(); return;
```

```
    }
    cout << "  Judul  : "; getline(cin, daftar_film[jumlah_film].judul_film);
    cout << "  Genre  : "; getline(cin, daftar_film[jumlah_film].genre_film);
    cout << "  Tahun  : "; cin >> daftar_film[jumlah_film].tahun_rilis;
cin.ignore();
    cout << "  Harga  : "; cin >> daftar_film[jumlah_film].harga_sewa;
cin.ignore();
    cout << "  Studio : "; getline(cin,
daftar_film[jumlah_film].studio.nama_studio);
    cout << "  Negara : "; getline(cin,
daftar_film[jumlah_film].studio.negara_studio);
    daftar_film[jumlah_film].tersedia = true;
    jumlah_film++;
    cout << "  [v] Film berhasil ditambahkan!" << endl;
    cout << "  [Tekan Enter...]"; cin.get();
}
```

### 3.10 Fitur Edit Film (Admin)

```
void menuEditFilm() {
    cout << "  === EDIT FILM ===" << endl;
    cout <<
"+-------+------------------------------+-----------+-------+----------+" <<
endl;
    cout << "| Kode  | Judul                        | Genre     | Tahun |
Harga/Hr |" << endl;
    cout <<
"+-------+------------------------------+-----------+-------+----------+" <<
endl;
    for (int i = 0; i < jumlah_film; i++) {
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << "| " << left << setw(5)  << daftar_film[i].kode_film
             << "| "         << setw(29) << daftar_film[i].judul_film
             << "| "         << setw(10) << daftar_film[i].genre_film
             << "| "         << setw(5)  << daftar_film[i].tahun_rilis
             << " | "        << setw(8)  << harga_str << "|" << endl;
    }
    cout <<
"+-------+------------------------------+-----------+-------+----------+" <<
endl;
    cin.ignore();
    string kode_diedit;
    cout << "  Kode film (Enter -> Kembali) : ";
    getline(cin, kode_diedit);
```

```
    if (kode_diedit == "") return;

    int indeks_edit = cariFilm(kode_diedit);
    if (indeks_edit == -1) {
        cout << "  [!] Kode tidak ditemukan." << endl; cin.get(); return;
    }
    cout << "  [1]Judul  [2]Genre  [3]Tahun  [4]Harga  [5]Studio  [6]Negara" <<
endl;
    cout << "  Pilihan  : "; int pilihan_edit; cin >> pilihan_edit;
cin.ignore();
    if      (pilihan_edit == 1) { cout << "  Judul baru  : "; getline(cin,
daftar_film[indeks_edit].judul_film); }
    else if (pilihan_edit == 2) { cout << "  Genre baru  : "; getline(cin,
daftar_film[indeks_edit].genre_film); }
    else if (pilihan_edit == 3) { cout << "  Tahun baru  : "; cin >>
daftar_film[indeks_edit].tahun_rilis; }
    else if (pilihan_edit == 4) { cout << "  Harga baru  : "; cin >>
daftar_film[indeks_edit].harga_sewa; }
    else if (pilihan_edit == 5) { cout << "  Studio baru : "; getline(cin,
daftar_film[indeks_edit].studio.nama_studio); }
    else if (pilihan_edit == 6) { cout << "  Negara baru : "; getline(cin,
daftar_film[indeks_edit].studio.negara_studio); }
    cout << "  [v] Film berhasil diedit!" << endl;
    cout << "  [Tekan Enter...]"; cin.get();
}
```

### 3.11 Fitur Hapus Film (Admin)

```
void menuHapusFilm() {
    cout << "  === HAPUS FILM ===" << endl;
    cout <<
"+-------+-----------------------------+-----------+-------+---------+" <<
endl;
    cout << "| Kode  | Judul                         | Genre     | Tahun | Status
|" << endl;
    cout <<
"+-------+-----------------------------+-----------+-------+---------+" <<
endl;
    for (int i = 0; i < jumlah_film; i++) {
        string status_film = daftar_film[i].tersedia ? "Tersedia" : "Disewa  ";
        cout << "| " << left << setw(5)  << daftar_film[i].kode_film
             << "| "         << setw(29) << daftar_film[i].judul_film
             << "| "         << setw(10) << daftar_film[i].genre_film
             << "| "         << setw(5)  << daftar_film[i].tahun_rilis
             << " | "        << status_film << "|" << endl;
    }
    cout <<
```

```
"+-------+-----------------------------+-----------+-------+---------+" <<
endl;
    cin.ignore();
    string kode_dihapus;
    cout << "  Kode film (Enter -> Kembali) : ";
    getline(cin, kode_dihapus);
    if (kode_dihapus == "") return;

    int indeks_hapus = cariFilm(kode_dihapus);
    if (indeks_hapus == -1) {
        cout << "  [!] Kode tidak ditemukan." << endl; cin.get(); return;
    }
    if (!daftar_film[indeks_hapus].tersedia) {
        cout << "  [!] Film sedang disewa, tidak bisa dihapus." << endl;
cin.get(); return;
    }
    cout << "  Hapus \"" << daftar_film[indeks_hapus].judul_film << "\"? (y/n) :
";
    char konfirmasi_hapus; cin >> konfirmasi_hapus;
    if (konfirmasi_hapus != 'y' && konfirmasi_hapus != 'Y') {
        cout << "  Dibatalkan." << endl; cin.ignore(); cin.get(); return;
    }
    for (int i = indeks_hapus; i < jumlah_film - 1; i++) daftar_film[i] =
daftar_film[i + 1];
    jumlah_film--;
    cout << "  [v] Film berhasil dihapus!" << endl;
    cout << "  [Tekan Enter...]"; cin.ignore(); cin.get();
}
```

### 3.12 Fitur Semua Riwayat (Admin)

```
void menuSemuaRiwayat() {
    cout << "  === SEMUA RIWAYAT SEWA ===" << endl;
    cout <<
"+----+-----------+-----------------------------+----------+---------+" <<
endl;
    cout << "| ID | Penyewa   | Film                        | Total    | Status
|" << endl;
    cout <<
"+----+-----------+-----------------------------+----------+---------+" <<
endl;
    if (jumlah_sewa == 0) cout << "  Belum ada riwayat." << endl;
    for (int i = 0; i < jumlah_sewa; i++) {
        string status_sewa  = daftar_sewa[i].sudah_kembali ? "Selesai " : "Aktif
";
        int    denda_tampil = daftar_sewa[i].sudah_kembali
                            ? daftar_sewa[i].denda
                            : hitungDenda(daftar_sewa[i].harga_per_hari, 0);
```

```
        int    total_tampil = daftar_sewa[i].total_bayar + denda_tampil;
        string total_str    = "Rp" + to_string(total_tampil);
        cout << "| " << left << setw(2)  << daftar_sewa[i].id_sewa
             << " | "        << setw(10) << daftar_sewa[i].nama_penyewa
             << "| "         << setw(29) << daftar_sewa[i].judul_film_sewa
             << "| "         << setw(8)  << total_str
             << "| "         << status_sewa << "|" << endl;
    }
    cout <<
"+----+-----------+-----------------------------+----------+---------+" <<
endl;
    cout << "  [Tekan Enter...]"; cin.ignore(); cin.get();
}
```

### 3.13 Main Program

```
int main() {
    inisialisasiData();

    bool program_berjalan = true;
    do {
        cout << "=============================================" << endl;
        cout << "         SELAMAT DATANG DI CINEMARENT        " << endl;
        cout << "  1. Login"                                    << endl;
        cout << "  2. Register"                                 << endl;
        cout << "  0. Keluar"                                   << endl;
        cout << "=============================================" << endl;
        cout << "  Pilihan : ";
        int pilihan_awal; cin >> pilihan_awal;

        if (pilihan_awal == 2) {
            prosesRegister("User");

        } else if (pilihan_awal == 1) {
            bool login_ok = prosesLogin(nama_aktif, password_aktif, role_aktif);

            if (login_ok) {
                bool sedang_login = true;
                do {
                    cout << "=============================================" <<
endl;
                    cout << "  MENU " << (role_aktif == "Admin" ? "ADMIN" :
```

```
"USER")
                        << " | " << nama_aktif << endl;
                    cout << "  1. Lihat Film"   << endl;
                    cout << "  2. Sewa Film"    << endl;
                    cout << "  3. Kembalikan"   << endl;
                    cout << "  4. Riwayat Sewa" << endl;
                    if (role_aktif == "Admin") {
                        cout << "-------------------------------------------" <<
endl;
                        cout << "  5. Tambah Film"   << endl;
                        cout << "  6. Edit Film"     << endl;
                        cout << "  7. Hapus Film"    << endl;
                        cout << "  8. Semua Riwayat" << endl;
                    }
                    cout << "  0. Logout" << endl;
                    cout << "-------------------------------------------" <<
endl;
                    cout << "  Pilihan : ";
                    int pilihan_menu; cin >> pilihan_menu;

                    switch (pilihan_menu) {
                        case 1: menuLihatFilm();
break;
                        case 2: menuSewaFilm();
break;
                        case 3: menuKembalikanFilm();
break;
                        case 4: menuRiwayatSewa(nama_aktif, password_aktif);
break;
                        case 5:
                            if (role_aktif == "Admin") menuTambahFilm();
                            else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                            break;
                        case 6:
                            if (role_aktif == "Admin") menuEditFilm();
                            else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                            break;
                        case 7:
                            if (role_aktif == "Admin") menuHapusFilm();
                            else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                            break;
                        case 8:
                            if (role_aktif == "Admin") menuSemuaRiwayat();
                            else { cout << "  [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                            break;
                        case 0:
                            cout << "  Sampai jumpa, " << nama_aktif << "!" <<
```

```
endl;
                            nama_aktif = ""; password_aktif = ""; role_aktif =
"";
                            sedang_login = false;
                            cout << "  [Tekan Enter...]"; cin.ignore();
cin.get();
                            break;
                        default:
                            cout << "  [!] Pilihan tidak valid." << endl;
                            cout << "  [Tekan Enter...]"; cin.ignore();
cin.get();
                            break;
                    }
                } while (sedang_login);
            }

        } else if (pilihan_awal == 0) {
            cout << "  Terima kasih! Sampai jumpa!" << endl;
            program_berjalan = false;
        } else {
            cout << "  [!] Pilihan tidak valid." << endl;
            cout << "  [Tekan Enter...]"; cin.ignore(); cin.get();
        }

    } while (program_berjalan);
    return 0;
}
```

## 4. Hasil Output

```
=============================================
        SELAMAT DATANG DI CINEMARENT
 1. Login
 2. Register
 0. Keluar
=============================================
  Pilihan :
```

Gambar 4.1 Menu Login

```
=============================================
        SELAMAT DATANG DI CINEMARENT
 1. Login
 2. Register
 0. Keluar
=============================================
  Pilihan : 1
  Nama     : Nana
  Password  : 009
  [!] Salah. Sisa percobaan: 2

  Nama     : Nala
  Password  : 008
  [!] Salah. Sisa percobaan: 1

  Nama     : Nurul
  Password  : 2509106044
  [!!!] Gagal 3 kali. Program berhenti.
```

Gambar 4.2 Login Gagal

```
=============================================
        SELAMAT DATANG DI CINEMARENT
 1. Login
 2. Register
 0. Keluar
=============================================
  Pilihan : 1
  Nama     : Nurul
  Password  : 2509106033
  [v] Selamat datang, Nurul!
  [Tekan Enter...]
```

Gambar 4.3 Login Berhasil

```
=============================================
        SELAMAT DATANG DI CINEMARENT
 1. Login
 2. Register
 0. Keluar
=============================================
  Pilihan : 0
  Terima kasih! Sampai jumpa!
PS C:\Users\LOQ\Documents\praktikum-apl\post-test\post-test-apl-2>
```

Gambar 4.4 Keluar

```
=============================================
        SELAMAT DATANG DI CINEMARENT
 1. Login
 2. Register
 0. Keluar
=============================================
  Pilihan : 2
  Nama     : Nana
  Password  : 009
  [v] Akun berhasil dibuat!
  [Tekan Enter...]
```

Gambar 4.5 Register

```
---------------------------------------------
 MENU USER | Nurul
 1. Lihat Film
 2. Sewa Film
 3. Kembalikan
 4. Riwayat Sewa
 0. Logout
---------------------------------------------
 Pilihan :
```

Gambar 4.6 Menu User

```
=============================================
  MENU USER | Nurul
  1. Lihat Film
  2. Sewa Film
  3. Kembalikan
  4. Riwayat Sewa
  0. Logout
---------------------------------------------
  Pilihan : 1
+------+----------------------------+-----------+------+----------+---------+
| Kode | Judul                      | Genre     | Tahun | Harga/Hr | Status  |
+------+----------------------------+-----------+------+----------+---------+
| F001 | The Nun II                 | Horror    | 2023 | Rp12000  | Tersedia|
| F002 | Hereditary                 | Horror    | 2018 | Rp12000  | Tersedia|
| F003 | It (2017)                  | Horror    | 2017 | Rp12000  | Tersedia|
| F004 | Talk to Me                 | Horror    | 2023 | Rp12000  | Tersedia|
| F005 | Your Name                  | Romance   | 2016 | Rp10000  | Tersedia|
| F006 | To All The Boys            | Romance   | 2018 | Rp10000  | Tersedia|
| F007 | Five Feet Apart            | Romance   | 2019 | Rp10000  | Tersedia|
| F008 | A Silent Voice             | Romance   | 2016 | Rp10000  | Tersedia|
| F009 | The Dark Knight            | Action    | 2008 | Rp13000  | Tersedia|
| F010 | John Wick                  | Action    | 2014 | Rp13000  | Tersedia|
| F011 | Mad Max: Fury Road         | Action    | 2015 | Rp13000  | Tersedia|
| F012 | Demon Slayer: Mugen        | Action    | 2020 | Rp13000  | Tersedia|
| F013 | Saltburn                   | Thriller  | 2023 | Rp12000  | Tersedia|
| F014 | Gone Girl                  | Thriller  | 2014 | Rp12000  | Tersedia|
| F015 | Bird Box                   | Thriller  | 2018 | Rp12000  | Tersedia|
| F016 | Midsommar                  | Thriller  | 2019 | Rp12000  | Tersedia|
| F017 | Super Mario Bros Movie     | Comedy    | 2023 | Rp10000  | Tersedia|
| F018 | Puss in Boots 2            | Comedy    | 2022 | Rp10000  | Tersedia|
| F019 | How To Train Your Dragon   | Animation | 2014 | Rp10000  | Tersedia|
| F020 | Spirited Away              | Animation | 2001 | Rp10000  | Tersedia|
+------+----------------------------+-----------+------+----------+---------+
  [Tekan Enter...]
```

Gambar 4.7 Menu User Pilihan 1

```
-----------------------------------------------
  Pilihan : 2
  === SEWA FILM ===
+-------+-----------------------------+-----------+----------+
| Kode  | Judul                       | Genre     | Harga/Hr |
+-------+-----------------------------+-----------+----------+
| F001 | The Nun II                   | Horror    | Rp12000 |
| F002 | Hereditary                   | Horror    | Rp12000 |
| F003 | It (2017)                    | Horror    | Rp12000 |
| F004 | Talk to Me                   | Horror    | Rp12000 |
| F005 | Your Name                    | Romance   | Rp10000 |
| F006 | To All The Boys              | Romance   | Rp10000 |
| F007 | Five Feet Apart              | Romance   | Rp10000 |
| F008 | A Silent Voice               | Romance   | Rp10000 |
| F009 | The Dark Knight              | Action    | Rp13000 |
| F010 | John Wick                    | Action    | Rp13000 |
| F011 | Mad Max: Fury Road           | Action    | Rp13000 |
| F012 | Demon Slayer: Mugen          | Action    | Rp13000 |
| F013 | Saltburn                     | Thriller  | Rp12000 |
| F014 | Gone Girl                    | Thriller  | Rp12000 |
| F015 | Bird Box                     | Thriller  | Rp12000 |
| F016 | Midsommar                    | Thriller  | Rp12000 |
| F017 | Super Mario Bros Movie       | Comedy    | Rp10000 |
| F018 | Puss in Boots 2              | Comedy    | Rp10000 |
| F019 | How To Train Your Dragon     | Animation | Rp10000 |
| F020 | Spirited Away                | Animation | Rp10000 |
+-------+-----------------------------+-----------+----------+
  Kode film (Enter -> Kembali) : F012
  Tanggal Sewa:
  Hari  (1-31) : 4
  Bulan (1-12) : 6
  Tahun        : 2025
  Durasi (hari, min 1) : 6
  Film         : Demon Slayer: Mugen
  Tgl Sewa     : 04/06/2025
  Wajib Kembali : 10/06/2025
  Total Bayar   : Rp78000
  Denda         : Rp5000/hari jika terlambat
  Konfirmasi? (y/n) : y
  [v] Berhasil menyewa "Demon Slayer: Mugen"!
  [Tekan Enter...]
```

Gambar 4.8 Menu User Pilihan 2

```
-----------------------------------------------
  Pilihan : 3
  === KEMBALIKAN FILM ===
+----+-----------------------------+------------+------------+
| ID | Judul                       | Tgl Sewa   | Wajib Kemb |
+----+-----------------------------+------------+------------+
| 1  | Demon Slayer: Mugen         | 04/06/2025 | 10/06/2025 |
+----+-----------------------------+------------+------------+
  ID sewa : 1
  Tanggal Kembali:
  Hari  (1-31) : 9
  Bulan (1-12) : 6
  Tahun        : 2025
  [v] Film berhasil dikembalikan!
  Tepat waktu, tidak ada denda.
  TOTAL BAYAR: Rp78000
  [Tekan Enter...]
```

Gambar 4.9 Menu User Pilihan 3

```
==============================================
  MENU USER | Nurul
  1. Lihat Film
  2. Sewa Film
  3. Kembalikan
  4. Riwayat Sewa
  0. Logout
----------------------------------------------
  Pilihan : 4
  === RIWAYAT SEWA - Nurul ===
+----+------------------------------+----------+---------+
| ID | Judul                        | Total    | Status  |
+----+------------------------------+----------+---------+
| 1  | Demon Slayer: Mugen          | Rp78000  | Selesai |
+----+------------------------------+----------+---------+
  [Tekan Enter...]
```

Gambar 4.10 Menu User Pilihan 4

```
==============================================
  MENU ADMIN | Admin
  1. Lihat Film
  2. Sewa Film
  3. Kembalikan
  4. Riwayat Sewa
----------------------------------------------
  5. Tambah Film
  6. Edit Film
  7. Hapus Film
  8. Semua Riwayat
  0. Logout
----------------------------------------------
  Pilihan :
```

Gambar 4.11 Menu Admin

```
----------------------------------------------
  Pilihan : 5
  === TAMBAH FILM ===
  Daftar film saat ini:
+------+----------------------------+----------+-------+---------+
| Kode | Judul                      | Genre    | Tahun | Harga/Hr |
+------+----------------------------+----------+-------+---------+
| F001 | The Nun II                 | Horror   | 2023 | Rp12000  |
| F002 | Hereditary                 | Horror   | 2018 | Rp12000  |
| F003 | It (2017)                  | Horror   | 2017 | Rp12000  |
| F004 | Talk to Me                 | Horror   | 2023 | Rp12000  |
| F005 | Your Name                  | Romance  | 2016 | Rp10000  |
| F006 | To All The Boys            | Romance  | 2018 | Rp10000  |
| F007 | Five Feet Apart            | Romance  | 2019 | Rp10000  |
| F008 | A Silent Voice             | Romance  | 2016 | Rp10000  |
| F009 | The Dark Knight            | Action   | 2008 | Rp13000  |
| F010 | John Wick                  | Action   | 2014 | Rp13000  |
| F011 | Mad Max: Fury Road         | Action   | 2015 | Rp13000  |
| F012 | Demon Slayer: Mugen        | Action   | 2020 | Rp13000  |
| F013 | Saltburn                   | Thriller | 2023 | Rp12000  |
| F014 | Gone Girl                  | Thriller | 2014 | Rp12000  |
| F015 | Bird Box                   | Thriller | 2018 | Rp12000  |
| F016 | Midsommar                  | Thriller | 2019 | Rp12000  |
| F017 | Super Mario Bros Movie     | Comedy   | 2023 | Rp10000  |
| F018 | Puss in Boots 2            | Comedy   | 2022 | Rp10000  |
| F019 | How To Train Your Dragon   | Animation | 2014 | Rp10000 |
| F020 | Spirited Away              | Animation | 2001 | Rp10000 |
+------+----------------------------+----------+-------+---------+
  Kode   : F021
  Judul  : My Neighbor Totoro
  Genre  : Comedy
  Tahun  : 1998
  Harga  : 12000
  Studio : Ghibli
  Negara : Jepang
  [v] Film ditambahkan!
  [Tekan Enter...]
----------------------------------------------
```

Gambar 4.12 Menu Admin Pilihan 5

```
---------------------------------------------
  Pilihan : 1
+-------+-----------------------------+-----------+-------+----------+---------+
| Kode  | Judul                       | Genre     | Tahun | Harga/Hr | Status  |
+-------+-----------------------------+-----------+-------+----------+---------+
| F001  | The Nun II                  | Horror    | 2023 | Rp12000  | Tersedia|
| F002  | Hereditary                  | Horror    | 2018 | Rp12000  | Tersedia|
| F003  | It (2017)                   | Horror    | 2017 | Rp12000  | Tersedia|
| F004  | Talk to Me                  | Horror    | 2023 | Rp12000  | Tersedia|
| F005  | Your Name                   | Romance   | 2016 | Rp10000  | Tersedia|
| F006  | To All The Boys             | Romance   | 2018 | Rp10000  | Tersedia|
| F007  | Five Feet Apart             | Romance   | 2019 | Rp10000  | Tersedia|
| F008  | A Silent Voice              | Romance   | 2016 | Rp10000  | Tersedia|
| F009  | The Dark Knight             | Action    | 2008 | Rp13000  | Tersedia|
| F010  | John Wick                   | Action    | 2014 | Rp13000  | Tersedia|
| F011  | Mad Max: Fury Road          | Action    | 2015 | Rp13000  | Tersedia|
| F012  | Demon Slayer: Mugen         | Action    | 2020 | Rp13000  | Tersedia|
| F013  | Saltburn                    | Thriller  | 2023 | Rp12000  | Tersedia|
| F014  | Gone Girl                   | Thriller  | 2014 | Rp12000  | Tersedia|
| F015  | Bird Box                    | Thriller  | 2018 | Rp12000  | Tersedia|
| F016  | Midsommar                   | Thriller  | 2019 | Rp12000  | Tersedia|
| F017  | Super Mario Bros Movie      | Comedy    | 2023 | Rp10000  | Tersedia|
| F018  | Puss in Boots 2             | Comedy    | 2022 | Rp10000  | Tersedia|
| F019  | How To Train Your Dragon    | Animation | 2014 | Rp10000  | Tersedia|
| F020  | Spirited Away               | Animation | 2001 | Rp10000  | Tersedia|
| F021  | My Neighbor Totoro          | Comedy    | 1998 | Rp12000  | Tersedia|
+-------+-----------------------------+-----------+-------+----------+---------+
  [Tekan Enter...]
```

Gambar 4.13 Hasil Penambahan

```
---------------------------------------------
  Pilihan : 6
  === EDIT FILM ===
+-------+-----------------------------+-----------+-------+----------+
| Kode  | Judul                       | Genre     | Tahun | Harga/Hr |
+-------+-----------------------------+-----------+-------+----------+
| F001  | The Nun II                  | Horror    | 2023 | Rp12000  |
| F002  | Hereditary                  | Horror    | 2018 | Rp12000  |
| F003  | It (2017)                   | Horror    | 2017 | Rp12000  |
| F004  | Talk to Me                  | Horror    | 2023 | Rp12000  |
| F005  | Your Name                   | Romance   | 2016 | Rp10000  |
| F006  | To All The Boys             | Romance   | 2018 | Rp10000  |
| F007  | Five Feet Apart             | Romance   | 2019 | Rp10000  |
| F008  | A Silent Voice              | Romance   | 2016 | Rp10000  |
| F009  | The Dark Knight             | Action    | 2008 | Rp13000  |
| F010  | John Wick                   | Action    | 2014 | Rp13000  |
| F011  | Mad Max: Fury Road          | Action    | 2015 | Rp13000  |
| F012  | Demon Slayer: Mugen         | Action    | 2020 | Rp13000  |
| F013  | Saltburn                    | Thriller  | 2023 | Rp12000  |
| F014  | Gone Girl                   | Thriller  | 2014 | Rp12000  |
| F015  | Bird Box                    | Thriller  | 2018 | Rp12000  |
| F016  | Midsommar                   | Thriller  | 2019 | Rp12000  |
| F017  | Super Mario Bros Movie      | Comedy    | 2023 | Rp10000  |
| F018  | Puss in Boots 2             | Comedy    | 2022 | Rp10000  |
| F019  | How To Train Your Dragon    | Animation | 2014 | Rp10000  |
| F020  | Spirited Away               | Animation | 2001 | Rp10000  |
| F021  | My Neighbor Totoro          | Comedy    | 1998 | Rp12000  |
+-------+-----------------------------+-----------+-------+----------+
  Kode film (Tekan Enter -> kembali) : F021
  [1]Judul [2]Genre [3]Tahun [4]Harga [5]Studio [6]Negara
  Pilihan : 3
  Tahun baru  : 1988
  [v] Film diedit!
  [Tekan Enter...]=============================================
```

Gambar 4.14 Menu Admin Pilihan 6

```
---------------------------------------------
 Pilihan : 1
+-------+----------------------------+-----------+-------+----------+---------+
| Kode  | Judul                      | Genre     | Tahun | Harga/Hr | Status  |
+-------+----------------------------+-----------+-------+----------+---------+
| F001  | The Nun II                 | Horror    | 2023  | Rp12000  | Tersedia|
| F002  | Hereditary                 | Horror    | 2018  | Rp12000  | Tersedia|
| F003  | It (2017)                  | Horror    | 2017  | Rp12000  | Tersedia|
| F004  | Talk to Me                 | Horror    | 2023  | Rp12000  | Tersedia|
| F005  | Your Name                  | Romance   | 2016  | Rp10000  | Tersedia|
| F006  | To All The Boys            | Romance   | 2018  | Rp10000  | Tersedia|
| F007  | Five Feet Apart            | Romance   | 2019  | Rp10000  | Tersedia|
| F008  | A Silent Voice             | Romance   | 2016  | Rp10000  | Tersedia|
| F009  | The Dark Knight            | Action    | 2008  | Rp13000  | Tersedia|
| F010  | John Wick                  | Action    | 2014  | Rp13000  | Tersedia|
| F011  | Mad Max: Fury Road         | Action    | 2015  | Rp13000  | Tersedia|
| F012  | Demon Slayer: Mugen        | Action    | 2020  | Rp13000  | Tersedia|
| F013  | Saltburn                   | Thriller  | 2023  | Rp12000  | Tersedia|
| F014  | Gone Girl                  | Thriller  | 2014  | Rp12000  | Tersedia|
| F015  | Bird Box                   | Thriller  | 2018  | Rp12000  | Tersedia|
| F016  | Midsommar                  | Thriller  | 2019  | Rp12000  | Tersedia|
| F017  | Super Mario Bros Movie     | Comedy    | 2023  | Rp10000  | Tersedia|
| F018  | Puss in Boots 2            | Comedy    | 2022  | Rp10000  | Tersedia|
| F019  | How To Train Your Dragon   | Animation | 2014  | Rp10000  | Tersedia|
| F020  | Spirited Away              | Animation | 2001  | Rp10000  | Tersedia|
| F021  | My Neighbor Totoro         | Comedy    | 1988  | Rp12000  | Tersedia|
+-------+----------------------------+-----------+-------+----------+---------+
 [Tekan Enter...]
```

Gambar 4.15 Hasil Pengeditan

```
 Pilihan : 7
 === HAPUS FILM ===
+-------+----------------------------+-----------+-------+---------+
| Kode  | Judul                      | Genre     | Tahun | Status  |
+-------+----------------------------+-----------+-------+---------+
| F001  | The Nun II                 | Horror    | 2023  | Tersedia|
| F002  | Hereditary                 | Horror    | 2018  | Tersedia|
| F003  | It (2017)                  | Horror    | 2017  | Tersedia|
| F004  | Talk to Me                 | Horror    | 2023  | Tersedia|
| F005  | Your Name                  | Romance   | 2016  | Tersedia|
| F006  | To All The Boys            | Romance   | 2018  | Tersedia|
| F007  | Five Feet Apart            | Romance   | 2019  | Tersedia|
| F008  | A Silent Voice             | Romance   | 2016  | Tersedia|
| F009  | The Dark Knight            | Action    | 2008  | Tersedia|
| F010  | John Wick                  | Action    | 2014  | Tersedia|
| F011  | Mad Max: Fury Road         | Action    | 2015  | Tersedia|
| F012  | Demon Slayer: Mugen        | Action    | 2020  | Tersedia|
| F013  | Saltburn                   | Thriller  | 2023  | Tersedia|
| F014  | Gone Girl                  | Thriller  | 2014  | Tersedia|
| F015  | Bird Box                   | Thriller  | 2018  | Tersedia|
| F016  | Midsommar                  | Thriller  | 2019  | Tersedia|
| F017  | Super Mario Bros Movie     | Comedy    | 2023  | Tersedia|
| F018  | Puss in Boots 2            | Comedy    | 2022  | Tersedia|
| F019  | How To Train Your Dragon   | Animation | 2014  | Tersedia|
| F020  | Spirited Away              | Animation | 2001  | Tersedia|
| F021  | My Neighbor Totoro         | Comedy    | 1988  | Tersedia|
+-------+----------------------------+-----------+-------+---------+
 Kode film (Tekan Enter -> kembali) : F021
 Hapus "My Neighbor Totoro"? (y/n) : y
 [v] Film dihapus!
 [Tekan Enter...]
```

Gambar 4.16 Menu Admin Pilihan 7

```
---------------------------------------------
  Pilihan : 1
+-------+----------------------------+-----------+-------+----------+---------+
| Kode  | Judul                      | Genre     | Tahun | Harga/Hr | Status  |
+-------+----------------------------+-----------+-------+----------+---------+
| F001  | The Nun II                 | Horror    | 2023  | Rp12000  | Tersedia|
| F002  | Hereditary                 | Horror    | 2018  | Rp12000  | Tersedia|
| F003  | It (2017)                  | Horror    | 2017  | Rp12000  | Tersedia|
| F004  | Talk to Me                 | Horror    | 2023  | Rp12000  | Tersedia|
| F005  | Your Name                  | Romance   | 2016  | Rp10000  | Tersedia|
| F006  | To All The Boys            | Romance   | 2018  | Rp10000  | Tersedia|
| F007  | Five Feet Apart            | Romance   | 2019  | Rp10000  | Tersedia|
| F008  | A Silent Voice             | Romance   | 2016  | Rp10000  | Tersedia|
| F009  | The Dark Knight            | Action    | 2008  | Rp13000  | Tersedia|
| F010  | John Wick                  | Action    | 2014  | Rp13000  | Tersedia|
| F011  | Mad Max: Fury Road         | Action    | 2015  | Rp13000  | Tersedia|
| F012  | Demon Slayer: Mugen        | Action    | 2020  | Rp13000  | Tersedia|
| F013  | Saltburn                   | Thriller  | 2023  | Rp12000  | Tersedia|
| F014  | Gone Girl                  | Thriller  | 2014  | Rp12000  | Tersedia|
| F015  | Bird Box                   | Thriller  | 2018  | Rp12000  | Tersedia|
| F016  | Midsommar                  | Thriller  | 2019  | Rp12000  | Tersedia|
| F017  | Super Mario Bros Movie     | Comedy    | 2023  | Rp10000  | Tersedia|
| F018  | Puss in Boots 2            | Comedy    | 2022  | Rp10000  | Tersedia|
| F019  | How To Train Your Dragon   | Animation | 2014  | Rp10000  | Tersedia|
| F020  | Spirited Away              | Animation | 2001  | Rp10000  | Tersedia|
+-------+----------------------------+-----------+-------+----------+---------+
  [Tekan Enter...]
```

Gambar 4.17 Hasil Penghapusan

```
  ==============================================
    MENU ADMIN | Admin
    1. Lihat Film
    2. Sewa Film
    3. Kembalikan
    4. Riwayat Sewa
  ------------------------------------------
    5. Tambah Film
    6. Edit Film
    7. Hapus Film
    8. Semua Riwayat
    0. Logout
  ----------------------------------------------
    Pilihan : 8
    === SEMUA RIWAYAT SEWA ===
  +----+-----------+----------------------------+----------+---------+
  | ID | Penyewa   | Film                       | Total    | Status  |
  +----+-----------+----------------------------+----------+---------+
  | 1  | Nurul     | Demon Slayer: Mugen        | Rp78000  | Selesai |
  +----+-----------+----------------------------+----------+---------+
    [Tekan Enter...]
```

Gambar 4.18 Menu Admin Pilihan 8

```
  ==============================================
    MENU ADMIN | Admin
    1. Lihat Film
    2. Sewa Film
    3. Kembalikan
    4. Riwayat Sewa
  ------------------------------------------
    5. Tambah Film
    6. Edit Film
    7. Hapus Film
    8. Semua Riwayat
    0. Logout
  ----------------------------------------------
    Pilihan : 0
    Sampai jumpa, Admin!
    [Tekan Enter...]
```

Gambar 4.19 Menu User dan Admin Pilihan 0

## 5. Langkah-langkah GIT

### 5.1 GIT Add

```
PS C:\Users\LOQ\Documents\praktikum-apl> git add .
```

Gambar 5.1 Git Add

Digunakan untuk menambahkan perubahan file ke staging area (area persiapan).

### 5.2 GIT Commit

```
PS C:\Users\LOQ\Documents\praktikum-apl> git add .
PS C:\Users\LOQ\Documents\praktikum-apl> git commit -m "Post-test 3 Codingan"
[main 6e51b7a] Post-test 3 Codingan
 6 files changed, 919 insertions(+)
 rename A2'25/{pertemuan-1 => Pertemuan-2}/Pertemuan-2.cpp (100%)
 rename A2'25/{pertemuan-1 => Pertemuan-2}/Pertemuan-2.exe (100%)
 create mode 100644 A2'25/Pertemuan-3/Pertemuan-3.cpp
 create mode 100644 A2'25/Pertemuan-3/Pertemuan-3.exe
 create mode 100644 post-test/post-test-apl-3/2509106033_NurulLatifa_PT-3.cpp
 create mode 100644 post-test/post-test-apl-3/2509106033_NurulLatifa_PT-3.exe
```

Gambar 5.2 Git Commit

Digunakan untuk menyimpan perubahan yang sudah di-add ke dalam repository lokal dengan pesan tertentu.

### 5.3 GIT Push

```
PS C:\Users\LOQ\Documents\praktikum-apl> git add .
PS C:\Users\LOQ\Documents\praktikum-apl> git commit -m "Post-test 3 Codingan"
[main 6e51b7a] Post-test 3 Codingan
 6 files changed, 919 insertions(+)
 rename A2'25/{pertemuan-1 => Pertemuan-2}/Pertemuan-2.cpp (100%)
 rename A2'25/{pertemuan-1 => Pertemuan-2}/Pertemuan-2.exe (100%)
 create mode 100644 A2'25/Pertemuan-3/Pertemuan-3.cpp
 create mode 100644 A2'25/Pertemuan-3/Pertemuan-3.exe
 create mode 100644 post-test/post-test-apl-3/2509106033_NurulLatifa_PT-3.cpp
 create mode 100644 post-test/post-test-apl-3/2509106033_NurulLatifa_PT-3.exe
PS C:\Users\LOQ\Documents\praktikum-apl> git push -u origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 20 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 90.16 KiB | 4.29 MiB/s, done.
Total 12 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   https://github.com/nulaatifa/praktikum-apl.git
To https://github.com/nulaatifa/Praktikum_Algoritma_Pemograman_Lanjut.git
   88067e2..6e51b7a  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\LOQ\Documents\praktikum-apl>
```

Gambar 5.3 Git Push

Digunakan untuk mengirim hasil commit dari repository lokal ke repository remote (misalnya ke GitHub).